

6th Grade Unit 3 Assessment Guide

General Information	The benchmarks assessed on this assessment may be challenging to some students. The questions on the assessment are designed to show teachers where students may have a weakness. The questions for MA.6.NSO.3.5 use a variety of numbers and forms so that teachers can note if students are struggling with a certain form. Some of the questions for MA.6.NSO.2.3 are purposefully scaffolded to help teachers determine if students struggle with discerning the necessary steps to complete a multi-step problem or the student struggles with operations with rational numbers. Unit 1 is designed to be a support for students as they transition from grade 5 to grade 6. If students struggled in both units 1 and 3, teachers may want to look for ways to support student learning and growth on real-world problems where students use rational numbers. Students may struggle with the percent form of numbers where the value is greater than 100 or includes a decimal. Recognizing equivalent forms of rational numbers helps students grow in their number sense. The benchmark, MA.6.NSO.3.5, does not state that students have to write rational numbers written as a fraction in “lowest terms” or “simplified”. The benchmark states that students should write rational numbers in “different, but equivalent” forms. The answers on this assessment do not require the student to write fractions in “lowest terms”. This is in alignment with the benchmark. The Grade 6 B.E.S.T. Instructional Guide for Mathematics states that the prerequisite benchmarks for MA.6.NSO.2.3 are MA.5.AR.1.1, MA.5.AR.1.2, MA.5.AR.1.3, MA.5.M.2.1, and MA.5.GR.2.1. Teachers who find that students struggle with multi-step problems may want to use the Grade 5 B.E.S.T. Instructional Guide for Mathematics to gain insight on the expectations of a grade 5 student as well as to find misconceptions and instructional tasks that can be used to support the struggling student.
Calculator Information	The assessment is no calculator.
Total Recommended Points	The total recommended points in this guide is 95 points. Teachers can either determine where best to use the extra 5 points or round the total value up to 100 points.

Question	1		Benchmark(s)	MA.6.NSO.3.5				
<table><tr><th>Decimal Form</th><th>Percent Form</th></tr><tr><td>9.375</td><td>937.5%</td></tr></table>			Decimal Form	Percent Form	9.375	937.5%	Recommended Scoring (6 Total Points)	Consider giving students 3 points for each equivalent rational number in the correct form.
Decimal Form	Percent Form							
9.375	937.5%							
Question	2		Benchmark(s)	MA.6.NSO.3.5				
<table><tr><th>Fraction Form</th><th>Decimal Form</th></tr><tr><td>$\frac{325}{1000}$</td><td>0.325</td></tr></table>			Fraction Form	Decimal Form	$\frac{325}{1000}$	0.325	Recommended Scoring (6 Total Points)	Consider giving students 3 points for each correct equivalent rational number in the correct form. Please note that students could write $\frac{325}{1000}$ as $\frac{65}{200}$ or $\frac{13}{40}$.
Fraction Form	Decimal Form							
$\frac{325}{1000}$	0.325							
Question	3		Benchmark(s)	MA.6.NSO.3.5				
<table><tr><th>Fraction Form</th><th>Percent Form</th></tr><tr><td>$\frac{73}{100}$</td><td>73%</td></tr></table>			Fraction Form	Percent Form	$\frac{73}{100}$	73%	Recommended Scoring (6 Total Points)	Consider giving students 3 points for each equivalent rational number in the correct form.
Fraction Form	Percent Form							
$\frac{73}{100}$	73%							

Question	4	Benchmark(s)	MA.6.NSO.3.5
	☐ 0.225 ☒ 2.25 ☐ 22.5 ☐ $\frac{225}{1000}$ ☒ $\frac{225}{100}$ ☐ $\frac{9}{40}$ ☒ $\frac{9}{4}$	Recommended Scoring (9 Total Points)	Consider giving students 3 points for each equivalent rational number in the correct form.
Question	5	Benchmark(s)	MA.6.NSO.3.5
	☐ 0.3264% ☒ 3264% ☒ $\frac{816}{25}$ ☐ $\frac{3264}{1000}$ ☒ $\frac{3264}{100}$ ☒ $32\frac{32}{50}$ ☐ $32\frac{1}{2}$	Recommended Scoring (12 Total Points)	Consider giving students 3 points for each equivalent rational number in the correct form.

Question	6a	Benchmark(s)	MA.6.NSO.2.3
112.422		Recommended Scoring (3 Total Points)	Consider giving partial credit based on student work. Students should not round the answer.
Question	6b	Benchmark(s)	MA.6.NSO.2.3
59		Recommended Scoring (4 Total Points)	Consider giving partial credit based on student work. Consider giving 1 point for correctly rounding to a whole number.
Question	7	Benchmark(s)	MA.6.NSO.2.3
Year	Average emissions of CO ₂ (in grams) per kilometer for 30 cars		Recommended Scoring (9 Total Points)
2001	5232½		
2019	3217½		
Students may also find the difference and then multiply by 30. 174½ - 107¼ = 67⅔ 30 · (67⅔) = 2,015			
Difference	2,015		
			Consider giving partial credit based on student work. If students first found the average emissions for 30 cars each year: • 3 points for 2001 • 3 points for 2019 • 3 points for the difference If students first found the difference: • 4.5 points for the difference of each year • 4.5 points for multiplying by 30

Question	8a	Benchmark(s)	MA.6.NSO.2.3
Cost of 1200 shares on 12/12/1980 \$120.72		Recommended Scoring (3 Total Points)	Questions 8a, 8b, and 8c are a scaffolded multi-step problem. Consider giving partial credit based on student work.
Question	8b	Benchmark(s)	MA.6.NSO.2.3
Cost of 1200 shares on 12/13/2010 \$11,902.08		Recommended Scoring (3 Total Points)	Questions 8a, 8b, and 8c are a scaffolded multi-step problem. Consider giving partial credit based on student work.
Question	8c	Benchmark(s)	MA.6.NSO.2.3
Difference in Cost \$11,781.36		Recommended Scoring (3 Total Points)	Questions 8a, 8b, and 8c are a scaffolded multi-step problem. Consider not penalizing if a student carries forward an error from question 8a or 8b but correctly demonstrates finding the difference in cost with the previous error.

Question	8d	Benchmark(s)	MA.6.NSO.2.3
Total Cost\$52.06		Recommended Scoring (9 Total Points)	Question 8d requires the student to do all of the steps without scaffolding. Consider giving partial credit based on student work. • 3 points for 12/12/1990 • 3 points for 12/12/2000 • 3 points for total cost
Question	8e	Benchmark(s)	MA.6.NSO.2.3
Amount For Each Child\$6076.25 or \$6076.26		Recommended Scoring (10 Total Points)	Question 8e requires the student to do all of the steps without scaffolding. Consider giving partial credit based on student work. • 3 points for 200(121.5251) • 3 points for $\frac{24305.02}{4}$ • 3 points for the correct answer • 1 point for rounding to hundredths based on the context.

Question	9	Benchmark(s)	MA.6.NSO.2.3
Feed Used (tons)	$\frac{3}{4}$	Recommended Scoring (12 Total Points)	Question 9 requires the student to do all of the steps without scaffolding. Consider giving partial credit based on student work. • 3 points for $\frac{37}{40} + \frac{1}{8}$ <i>If students used $\frac{5}{7}$:</i> • 3 points for determining that the amount used was $\frac{5}{7}$ • 3 points for $\frac{5}{7} \left(\frac{42}{40} \right)$ <i>If students used $\frac{2}{7}$:</i> • 3 points for $\frac{2}{7} \left(\frac{42}{40} \right)$ • 3 points for determining they need to subtract the result from $\frac{42}{40}$ • 3 points for the correct answer